Mid-term examination on Logic Circuit Design

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1 Binary numbers

Question. If $N = b_{n-1}b_{n-2}\dots b_0$ is an n-bit binary number, what is $2N$ and $\lfloor N/2 \rfloor$?

Answer. By definition

$$\begin{aligned}
N &= b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_1 2 + b_0\\
&= 2(b_{n-1}2^{n-2} + b_{n-2}2^{n-3} + \cdots + b_1) + b_0\\
\lfloor N/2 \rfloor &= b_{n-1}2^{n-2} + b_{n-2}2^{n-3} + \cdots + b_1\\
&= 0 \times 2^{n-1} + b_{n-1}2^{n-2} + b_{n-2}2^{n-3} + \cdots + b_1\\
&= 0b_{n-1}b_{n-2}\dots b_1
\end{aligned}$$

And

$$\begin{aligned}
2N &= 2(b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_1 2 + b_0)\\
&= b_{n-1}2^{n} + b_{n-2}2^{n-1} + \cdots + b_1 2^2 + b_0 2^1 + 0 \times 2^0\\
&= b_{n-1}b_{n-2}\dots b_1 b_0 0
\end{aligned}$$

2 Boolean algebra

Question. Using truth tables, prove the distributive laws

$$\begin{aligned}
A + BC &\stackrel{1}{=} (A+B)(A+C)\\
A(B+C) &\stackrel{2}{=} AB + AC
\end{aligned}$$

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\end{aligned}$$

A	B	C	BC	$\mathbf{A+BC}$	$A+B$	$A+C$	$(\mathbf{A+B})(\mathbf{A+C})$
0	0	0	0	**0**	0	0	**0**
0	0	1	0	**0**	0	1	**0**
0	1	0	0	**0**	1	0	**0**
0	1	1	1	**1**	1	1	**1**
1	0	0	0	**1**	1	1	**1**
1	0	1	0	**1**	1	1	**1**
1	1	0	0	**1**	1	1	**1**
1	1	1	1	**1**	1	1	**1**

A	B	C	$B+C$	$\mathbf{A(B+C)}$	AB	AC	$\mathbf{AB+AC}$
0	0	0	0	**0**	0	0	**0**
0	0	1	1	**0**	0	0	**0**
0	1	0	1	**0**	0	0	**0**
0	1	1	1	**0**	0	0	**0**
1	0	0	0	**0**	0	0	**0**
1	0	1	1	**1**	0	1	**1**
1	1	0	1	**1**	1	0	**1**
1	1	1	1	**1**	1	1	**1**

Given the distributive laws and the following

$$1+A \overset{3}{=} 1 \qquad 0 \cdot A \overset{4}{=} 0 \qquad 0+A \overset{5}{=} A \qquad 1 \cdot A \overset{6}{=} A$$
$$A+A \overset{7}{=} A \qquad A \cdot A \overset{8}{=} A \qquad A+\overline{A} \overset{9}{=} 1 \qquad A \cdot \overline{A} \overset{10}{=} 0 \qquad \overline{\overline{A}} \overset{11}{=} A$$

prove the absorption laws

$$A+AB \overset{12}{=} A$$
$$A(A+B) \overset{13}{=} A$$

without truth tables.

Question. Given the distributive laws and the following

$$1+A \overset{3}{=} 1 \qquad 0 \cdot A \overset{4}{=} 0 \qquad 0+A \overset{5}{=} A \qquad 1 \cdot A \overset{6}{=} A$$
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prove the absorption laws

$$A+AB = A \tag{1}$$
$$A(A+B) = A \tag{2}$$

without truth tables.

Question. Given the distributive laws and the following

$$1 + A \overset{3}{=} 1 \qquad 0 \cdot A \overset{4}{=} 0 \qquad 0 + A \overset{5}{=} A \qquad 1 \cdot A \overset{6}{=} A$$
$$A + A \overset{7}{=} A \qquad A \cdot A \overset{8}{=} A \qquad A + \overline{A} \overset{9}{=} 1 \qquad A \cdot \overline{A} \overset{10}{=} 0 \qquad \overline{\overline{A}} \overset{11}{=} A$$

prove the absorption laws

$$A + AB = A \tag{3}$$
$$A(A + B) = A \tag{4}$$

without truth tables.

3 Addition of BCD numbers

Question. State any overflow or underflow.

$$\text{(a)}\quad \begin{array}{r} 0111 \;\; 0101 \;\; 1001 \;\; 0000 \\ +\,0010 \;\; 1001 \;\; 0011 \;\; 0101 \\ \hline \end{array}$$

$$\text{(b)}\quad \begin{array}{r} 0011 \;\; 0110 \;\; 0001 \;\; 0111 \\ +\,0001 \;\; 0100 \;\; 0110 \;\; 1000 \\ \hline \end{array}$$

$$\text{(c)}\quad \begin{array}{r} 1001 \;\; 0101 \;\; 1000 \;\; 0111 \\ +\,0011 \;\; 0101 \;\; 0110 \;\; 0010 \\ \hline \end{array}$$

Answer. Note that there cannot be underflows since we are adding positive numbers. There cannot be overflows because the number of digits of the result is not specified.

(a)		1 1	1	1 1	
		0 1 1 1	0 1 0 1	1 0 0 1	0 0 0 0
		0 0 1 0	1 0 0 1	0 0 1 1	0 1 0 1
		1 1	1 1 1	1	
		1 0 0 1	1 1 1 0	1 1 0 0	0 1 0 1
			0 1 1 0	0 1 1 0	
	1	1 1			
		1 0 1 0	0 1 0 1	0 0 1 0	0 1 0 1
		0 1 1 0			
	0 0 0 1	0 0 0 0	0 1 0 1	0 0 1 0	0 1 0 1

(b)		1 1	1		
		0 0 1 1	0 1 1 0	0 0 0 1	0 1 1 1
		0 0 0 1	0 1 0 0	0 1 1 0	1 0 0 0
		1	1 1	1 1 1 1	1 1
		0 1 0 0	1 0 1 0	0 1 1 1	1 1 1 1
			0 1 1 0		0 1 1 0
		0 1 0 1	0 0 0 0	1 0 0 0	0 1 0 1

(c)		1 1	1 1		1 1
		1 0 0 1	0 1 0 1	1 0 0 0	0 1 1 1
		0 0 1 1	0 1 0 1	0 1 1 0	0 0 1 0
	1	1 1	1 1 1	1 1	
		1 1 0 0	1 0 1 0	1 1 1 0	1 0 0 1
		0 1 1 0	0 1 1 0	0 1 1 0	
	0 0 0 1	0 0 1 1	0 0 0 1	0 1 0 0	1 0 0 1